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23 November 1984
SUPERSEDING
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(See 6.7)

MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, MAIN CONDENSER CIRCULATING, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements applicable to the design and construction of centrifugal pumps for main condenser circulating water service on board Naval ships.

1.2 Classification. Pumps shall be of the following types, as specified (see 6.2.1):

Type I - Axial flow (propeller type).
Type II - Mixed flow.
Type III - Radial flow.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearings, Roller, Tapered.
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-C-465	- Copper-Aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630, 632M and 642); Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar), Shapes, and Forgings.
QQ-N-281	- Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections.
QQ-N-286	- Nickel-Copper-Aluminum Alloy, Wrought.
TT-P-645	- Primer, Paint, Zinc-Chromate, Alkyd Type.
GGG-E-950	- Extractor, Stuffing Box and Pump Packing (and Tamper); and Extractor, Lantern Gland.
GGG-P-781	- Puller, Mechanical Pull Attachment, Mechanical, and Puller Set, Mechanical.

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
DOD-D-1000	- Drawings, Engineering and Associated Lists.
MIL-S-1222	- Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-C-2212	- Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-S-8660	- Silicone Compound, Nato Code Number S-736.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-S-15083	- Steel Castings.
MIL-E-15090	- Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-15137	- Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
DOD-P-15328	- Primer (Wash), Pretreatment (Formula No. 117 for Metals) (Metric).
MIL-C-15345	- Castings, Nonferrous, Centrifugal.
MIL-C-15726	- Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar and Plate) and Forgings.
MIL-P-16789	- Packaging of Pumps, Including Prime Movers and Associated Repair Parts.
MIL-M-17060	- Motors, 60 Hertz, Alternating-Current, Integral Hp.
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.

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MILITARY (Continued)

MIL-M-17413 - Motors, Direct-Current, Integral Hp (Shipboard Use).
MIL-T-17523 - Turbine, Steam, Auxiliary (and Reduction Gear) Mechanical Drive.
DOD-P-17545 - Primer Coating, Alkyd-Red Lead Type, Formula No. 116 and Formula No. 116D (Metric).
MIL-B-17931 - Bearings, Ball, Annular, for Quiet Operation.
MIL-C-20159 - Copper-Nickel Alloy (70-30 and 90-10): Castings.
MIL-E-21562 - Electrodes and Rods - Welding, Bare, Nickel Alloy l
MIL-E-22200/3 - Electrodes, Welding, Covered: Nickel Base Alloys; and Cobalt Base Alloy.
MIL-S-22473 - Sealing, Locking, and Retaining Compounds; (Single-Component).
MIL-C-23233 - Couplings for Propulsion Units, Auxiliary Turbines and Line Shafts Naval Shipboard.
MIL-N-24106 - Nickel-Copper Alloy Bars, Rods, and Forgings.
MIL-P-24337 - Packing Material, Asbestos, Braided, Impregnated with PTFE (Polytetrafluoroethylene) , Surface Lubricated.
DOD-G-24508 - Grease, High Performance, Multi-Purpose (Metric).
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-H28/2 - Unified Thread Form and Thread Series for Bolts , Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Submarine Service.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
DOD-STD-2147 - Technical Repair Standards (TRS); Hull, Mechanical and Electrical (HM&E), Preparation of (Metric).

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MILITARY (Continued)

MS16142 - BOSS, Gasket Seal, Straight Thread Tube Fittings, Standard Dimensions for.
MS18116 - Bolt, Stud, and Socket Cap Head Screw, Nickel-Copper-Aluminum Alloy.
MS18229 - Plug for "O" Ring Gasket.

2.1.2 Government drawings and publications. The following Government drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND

B-153 - Packings and Gaskets, Standard Application of.
B-214 - Root Connections for Attaching Piping.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.
0900-LP-003-8000 - Metals, Surface Inspection Acceptance Standards.
0941-LP-041-3010 - Submarine Safety Design Review Procedures Manual.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

A 193 - Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service. (DoD adopted)
A 194 - Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service. (DoD adopted)
A 743 - Corrosion-Resistant Iron-Chromium, Iron-Chromium-Nickel, and Nickel-Base Alloy Castings for General Application. (DoD adopted)
A 744 - Corrosion-Resistant Iron-Chromium, Iron-Chromium-Nickel, and Nickel-Base Alloy Castings for Severe Service. (DoD adopted)
B 164 - Nickel-Copper Alloy Rod and Bar. (DoD adopted)
B 369 - Copper-Nickel Alloy Castings.
B 443 - Nickel-Chromium-Molybdenyum-Columbium Alloy (UNS N06625) Plate, Sheet, and Strip.

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ASTM (Continued)

B 446 - Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625) Rod and Bar.

B 463 - Chromium-Nickel-Iron-Molybdenum-Copper-Columbium Stabilized Alloy (UNS N08020) Plate, Steel, and Strip.

B 473 - Chromium-Nickel-Iron-Molybdenum-Copper-Columbium Stabilized Alloy (UNS N08020) Bar and Wire.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Test Code of Standards - Centrifugal Pump Section.

(Application for copies should be addressed to Hydraulic Institute, 712 Lakewood Center N., 14600 Detroit Avenue, Cleveland) OH 44107.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. A sample shall be subjected to first article inspection (see 4.3 and 6.3). On board repair parts and stocked spare parts ordered for pumps built to this specification shall be subject to the same tests as the original equipment.

3.2 General design. The design and construction of pumps shall comply with the space and weight requirements as specified (see 6.2.1). The pump unit shall perform under the list, pitch, roll and trim conditions as specified (see 6.2.1). Pumps shall operate continuously at any point over the entire flow range and the design conditions as specified (see 6.2.1).

3.2.1 Reliability. The requirements of this specification are imposed exclusively to obtain equipment of utmost reliability for the service intended. The assurance of maximum reliability shall be the paramount controlling principle in the design, fabrication, assembly, and testing of this equipment. Users of and contractors to this specification are urged to communicate to the Naval Sea Systems Command (NAVSEA) any findings related to the requirements or lack of requirements of this specification whereby improvement of equipment reliability can be achieved. On board repair parts and stocked spare parts ordered for pumps built to this specification shall be subject to the same tests as the original equipment.

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3.2.2 The equipment specified herein will be operated, maintained, and repaired on board Navy ships. Attention is directed to the high equipment density of shipboard machinery spaces, to the fact that maintenance and repair will be made underway in heavy seas, that equipment will operate unattended, and that maintenance personnel may not be seasoned mechanics. The requirements for maximum reliability directly relates to those shipboard environmental and service conditions, and they shall be fully considered in the pump design. The aspects of "human engineering" shall be fully explored and considered in the equipment design to minimize the possibility of failure through improper operation and maintenance, and to preclude personnel safety hazards. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable functionally and from considerations of strength.

3.2.2.1 The equipment shall have maximum repair accessibility for ease of examination of wearing parts and for simplicity of disassembly and proper reassembly. Positioning and alinement of parts in assembly shall employ positive means such as shoulder, tongue, and groove or other locating techniques whereby correct reassembly is repeatedly assured. In consonance with the concept of ease of maintenance and where not already specified, the pump designer shall consider all of the following features for incorporation in the design and technical documentation wherever such incorporation will significantly contribute to ease of maintenance without compromising performance reliability:

(a) Lifting lugs.
(b) Casing assembly guide pins.
(c) Alinement and positioning dowels.
(d) Jacking screws.
(e) Ease of access to couplings.
(f) Re-makeable piping connections.
(g) Piping out of way of maintenance access.
(h) Ease of packing, mechanical seal, bearing, and wearing ring replacement.
(i) Split casing, bearing housings, etc., (except for submarine service).
(j) Guards for rotating and moving components.
(k) Warning plates to prevent casualties to equipment and personnel.
(l) Provisions for connecting instruments for performance evaluation.
(m) Attached instruments for monitoring performance.
(n) Visibility and access to attached instruments.
(o) Thoroughness of operating instructions.
(p) Thoroughness of preventive maintenance instructions.
(q) Explicit assembly and disassembly instructions.
(r) Exploded views of critical assemblies.
(s) Adequate system diagrams.

3.2.2.2 Each pump shall have a service life of 30 years of which 40 percent shall be actual operation. There shall be no limit on the number of starts during the life of the pump. It shall be assumed that during the life of the pump, parts subject to unavoidable wear and deterioration (with the exception of packing and seals) will be replaced at intervals no shorter than 3 years. The parts subject to wear, deterioration, and normally requiring replacement at

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3-year intervals during the service life of the pump (with the exception of packing and seals) shall have a life of 10,000 actual operating hours, and they shall be identified in appropriate drawings and manuals. The requirement set forth above shall not be construed or interpreted as a warranty requirement, nor otherwise affect the manufacturer's warranty.

3.2.2.3 In the design of sea-connected pumps for submarine application, full consideration shall be given to cyclic nature of the pressure loading. The design shall be in accordance with NAVSEA 0941-LP-041-3010. Justification of design to the command or agency concerned shall be made by experimental stress analysis (see 4.6.7.5).

3.2.3 Shock and vibration.

3.2.3.1 Pumps and drivers, all appurtenances, and controls shall pass a shock test in accordance with MIL-S-901, grade A, as specified in 4.6.7.2.

3.2.3.2 Unless otherwise specified (see 6.2.1), shock mounts (resilient mounts for shock attenuation) shall not be used. When noise attenuation mounts are specified, the pump shall pass the specified shock test on the noise attenuation mounts.

3.2.3.3 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be not more than 1/32 inch larger than the bolts for sizes up to and including 3/4 inch, and no more than 1/16 inch larger than bolts of greater than 3/4 inch size. Mounting bolts for fastening the equipment shall conform to grade 2 or better of MIL-S-1222.

3.2.3.4 Pumps shall not be damaged or cause to malfunction by either internally excited vibrations or by the environmental vibrations. Units furnished under this specification shall be tested to meet the requirements of MIL-STD-167-1, type I and II (see 4.6.7.4).

3.2.4 Alinement.

3.2.4.1 The design of all pumps shall be such that alinement will not be disturbed or undue stresses set up in any part by normal vibration or contraction and expansion of piping attached thereto in service.

3.2.4.2 Alinement between separate components of a unit shall be maintained by means of keyways, rabbeted or tongue and groove joints, fitted bolts, or other means. Parts manufactured to the same drawings shall be interchangeable.

3.2.4.3 In no case shall a piece of equipment be rigidly supported from more than one plane.

3.2.5 Noise. The design, construction, and workmanship of the equipment shall be such that noise levels during operation at the design conditions will meet the airborne noise and structureborne vibration noise requirements as specified (see 6.2.1 and 4.6.4).

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3.2.6 Mounting.

3.2.6.1 Pump units shall be horizontally or vertically mounted, as specified (see 6.2.1).

3.2.6.2 Each horizontal flexible coupled pump and separable driving unit, complete with all appurtenances, shall be mounted on a common bedplate.

3.2.6.3 Bedplates shall be sufficiently rigid to permit handling, shipment, and installation of the units on board ship, to minimize misalinement of the assembled units; and such that normal distortion, weaving or vibration of the supporting structures on board ships cannot cause misalinement between the pumps and driving units.

3.2.6.4 Bearing and seating surfaces of bedplates shall be finish machined.

3.2.6.5 Each component part of an assembled flexible coupled unit supported directly by a bedplate shall be doweled thereto to facilitate reassembly and maintenance of alinement, except that final doweling of the driver shall be performed by the installing activity after final installation. The manufacturer shall furnish a warning plate indicating the above.

3.2.6.6 Vertical pumps shall be base mounted or center of gravity mounted. The base and support bracket shall have provisions for bolting to a foundation. Pumps which are resiliently mounted shall be supported as close to the center of gravity as practical.

3.2.7 Bearings. Suitable bearings shall be installed in each pump unit for counteracting (a) any unbalanced hydraulic or mechanical thrust in either direction and (b) the fact that rolling, pitching, listing and trim, and the athwartships, fore and aft, and the vertical distance from the pump to the ship's rolling and pitching center may introduce loads even though the unit is in hydraulic balance (see 6.2.1). The contractor shall calculate the resulting static and dynamic loads on the bearings and certify that the bearings are suitable for this loading (see 6.2.2).

3.2.7.1 Radial bearings.

3.2.7.1.1 Radial bearings may be of the hydrodynamic or rolling contact type. If specifically approved by the design review agency (see 6.4), hydrostatic bearings may be used.

3.2.7.1.2 Except as otherwise approved by the design review agency, rolling contact bearings shall be of the ball type in accordance with FF-B-171. Roller bearings may be recommended if use of ball bearings is impractical. If approved, roller bearings shall be in accordance with FF-B-185 or FF-B-187. Rolling contact bearings shall be selected to result in a minimum B-10 life of 10,000 hours calculated in accordance with FF-B-171, FF-B-185, or FF-B-187.

3.2.7.1.3 Rolling contact bearings specifically selected for quiet operation in accordance with MIL-B-17931 shall be used for pumps for submarine application and for surface ships where required to meet special noise requirements

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3.2.7.1.4 Hydrodynamic radial bearings may be of the fixed or self-alining type and shall operate under full-fluid film lubrication for all operating conditions, except boundary lubrication will be permitted during pump start-up and shut-down. Where the requirement for full-fluid film operation cannot be met, a hydrostatic bearing may be recommended by the contractor for design review agency approval. The hydrostatic bearing shall operate without metal-to-metal contact over the entire range of the pump characteristic curve.

3.2.7.1.5 Water lubricated hydrodynamic radial bearings shall not be provided with lubrication grooves unless specifically recommended by the contractor and approved by the design review agency. If a hydrodynamic bearing design having axial grooves is recommended by the contractor, positive proof of the direction of radial loading shall be established. The orientation of the bearing shall be such that the effective bearing arc will not be reduced to less than 120 degrees at any operating condition of the pump. Orientation of the bearing shall be maintained by positive means. The effective arc is defined herein as the continuous load-carrying surface which is symmetrical with respect to the load vector. Release for manufacture shall be subject to verification of the applied load direction, by experimental data, and derived from tests of manufacturer's own similar designs, Where the method or experimental data are considered inconclusive by the design review agency, the recommendation for a grooved bearing design will be disapproved.

3.2.7.1.6 The hydrodynamic and hydrostatic bearing surface and the journal shall be finished to 32 roughness height rating (RHR) or better.

3.2.7.1.7 Type III pumps utilizing water-lubricated bearings shall use one or more of the following means to minimize radial loading on the shaft:

(a) Double volute casing.
(b) Concentric casing.
(c) Diffusers.
(d) Modified concentric casing.

For modified concentric casings, the casing shall be circular over at least 270 degrees of the arc beginning at the cut water and proceeding in the direction of impeller rotation toward the discharge nozzle. For both concentric and modified concentric casings, the ratio of the casing diameter at the cut water to the impeller diameter shall be 1.3 or greater.

3.2.7.2 Thrust bearings.

3.2.7.2.1 A suitable thrust bearing shall be installed in each flexible coupled unit for counteracting the forces resulting from the conditions stated in 3.2.7.

3.2.7.2.1.1 Thrust bearings shall be of the combined radial and thrust rolling contact type in accordance with 3.2.7.1.2.

3.2.7.2.2 For rigidly coupled units, the thrust bearing for the unit shall be in the driver and shall comply with 3.2.7.2.1.1. For rigidly coupled units where the driver is a turbine rated 500 horsepower (hp) or over, the thrust bearing shall be as specified in MIL-T-17523.

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3.2.7.3 Installation.

3.2.7.3.1 Ball or roller bearings, when used, shall be installed in housings separate from and independent of the stuffing boxes and with adequate protection from gland leakage. In vertical pumps, rolling contact bearings, where used, shall be installed external to the pump between the coupling and gland seal. Means shall be provided to prevent escape of lubricant around the shaft. All baffles, wipers, and related parts shall be readily renewable.

3.2.7.3.2 Bearing brackets shall be either cast integral with the casing, securely bolted to the casing on a machined shoulder, or held in alinement with the casing by at least two heavy dowels and securely bolted thereto. The use of bolts alone for securing brackets is prohibited.

3.2.7.3.3 Bearing housings shall be cast integral with the bracket or secured thereto in such a manner as to insure alinement. Bearing housing to bracket seats shall be machined such that precision of bearing alinement is insured.

3.2.7.3.4 On axially-split-casing pumps, the shells for hydrodynamic and hydrostatic bearings shall be split along the axis and so arranged as to permit renewal or refitting without removal of the pump rotor from the casing. They shall be neatly fitted in their seating to prevent axial play.

3.2.8 Lubrication.

3.2.8.1 The lubrication of rolling contact bearing shall be in accordance with FF-B-171, FF-B-185, or FF-B-187, as applicable.

3.2.8.1.1 Rolling contact bearing lubrication method shall prevent bearing failure resulting from over-greasing. Bearing relubrication procedures shall be accomplished with the unit secured.

3.2.8.2 Rolling contact bearings having a DN value in excess of 200,000 shall be oil lubricated unless the contractor can demonstrate to the satisfaction of the design review agency that the specific grease lubrication bearing application has been proven in previous Navy shipboard service. Grease lubrication is preferred where DN is less than 200,000.

Where:

D = Diameter of bearing bore in millimeters,
N = Revolutions per minute (r/rein).

3.2.8.3 Where forced feed lubrication is provided for turbine-driven pumps, the lubricating system shall be in accordance with requirements of the applicable turbine specification (see 3.2.13).

3.2.8.4 Grease lubricants shall be in accordance with DOD-G-24508. Oil lubricants shall be in accordance with MIL-L-17331, military symbol 2190-TEP. The design shall be such that any qualified grease under DOD-G-24508 or oil under MIL-L-17331, military symbol 2190-TEP, can be used interchangeably with any other qualified product to the same specification. Bearings shall be capable of being relubricated without removing the bearings, and shall not require frequent lubrication if the pump is not operated over an extended period of time. Bearing relubrication procedures shall be accomplished with the unit secured.

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3.2.8.5 Thrust bearing housings shall be so designed that the thrust bearing will be adequately lubricated immediately when the shaft starts to turn. This shall be accomplished by properly locating lubricant supply and drain fittings, and by adequate sealing of the bearing housings so that, regardless of the length of time the units are idle, an adequate supply of lubricant will remain in the bearing housings.

3.2.9 Packing and gaskets. Packing and gaskets shall be in accordance with Drawing B-153. Packing shall be of the polytetrafluoroethylene impregnated type in accordance with MIL-P-24377. Gasket compression and tolerances shall not affect bearing bushing or wearing ring fits to an extent that will adversely affect reliability and performance. Spiral wound gaskets shall be used only where use of a sheet gasket or O-ring is impracticable. Internal spiral wound gaskets shall be completely enclosed to prevent the gasket from unwinding and entering the fluid system.

3.2.10 Threaded fasteners. Threaded parts, such as bolts, studs, and nuts, shall conform to FED-STD-H28, FED-STD-H28/2, and MIL-S-1222. The use of tap-bolts or cap-screws is prohibited unless explicitly discussed and justified in a submittal to the design review agency and approved by the design review agency and where the use of through bolts or studs is impractical. The setting end of the studs shall be a class 3 fit with anaerobic locking compounds in accordance with MIL-S-22473, type AV. The nut end shall be a class 3 fit. The recommendations of the anaerobic locking compound manufacturer regarding the use of a primer shall be adhered to (see 3.6.1.2.3.3). Studs installed using locking anaerobic compound shall be inspected with a procedure proposed by the pump manufacturer and approved by the design review agency. The inspection method shall be shown on the pump drawing.

3.2.10.1 The length of minimum stud engagement shall be computed as specified in FED-STD-H28.

3.2.10.2 Unless otherwise specified (see 6.2.1), threaded fasteners of normally stocked lengths shall be used. Male threads on threaded fasteners, after being installed and tightened, shall protrude at least one thread beyond the top of the nut or plastic locking element. Excessive protrusion shall be avoided, particularly where necessary clearances, accessibility, and safety are important. Where practical, the number of threads protruding shall not exceed five; however, in no case shall thread protrusion exceed ten threads. Washers shall not be used for the sole purpose of lessening thread protrusion.

3.2.11 Piping root connections. Root connections for attaching pipes shall be in accordance with Drawing B-214.

3.2.12 Welding and brazing. Welding shall be in accordance with MIL-STD-278. The use of silver brazed joints is prohibited for any joint which could be heated by future weld repair of the casing. Where silver joints are not prohibited, such joints shall be in accordance with NAVSEA 0900-LP-001-7000, The use of silver-brazed joints is prohibited in the fluid boundary for sea water pumps for submarine applications.

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3.2.13 Turbines.

3.2.13.1 Turbines for driving pumps shall be in accordance with MIL-T-17523 and as specified (see 6.2.1).

3.2.14 Motors. Drive motors shall have sealed insulation and shall be of the type and characteristics specified (see 6.2.1) and shall conform to MIL-M-17413 for direct current (d.c.) motors, and to MIL-M-17060 for alternating current (a.c.) motors. For surface ships applications, the motors shall be totally enclosed fan cooled. For submarines only, see 6.2.1 for complete motor specification.

3.2.14.1 For type I pumps, the hp rating shall be determined by the following formula. For types II and 111 pumps, the hp rating shall be such that the motor is not overloaded anywhere on the head-capacity curve for type I pumps.

$$\text{Motor hp} \geq \frac{1.05 \times \text{maximum water hp output of pump at specified conditions}}{\text{Efficiency of pump at point of maximum hp}}$$

3.2.14.2 The actual motor rating shall be in accordance with Navy standard motor ratings.

3.2.15 Controllers. Motor controllers shall be in accordance with MIL-C-2212 for a.c. and d.c. and shall have the characteristics as specified (see 6.2.1).

3.2.16 Painting. External unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment conforming to DOD-P-15328, and one coat of zinc chromate primer conforming to TT-P-645, or one coat of red lead primer conforming to DOD-P-17545, followed by a finish coat of light gray equipment enamel conforming to MIL-E-15090.

3.2.16.1 Painting of external surfaces of nonferrous parts of pumps will not be required.

3.2.17 Identification plates.

3.2.17.1 Identification plates shall be furnished on each pump and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that identification plates of plastic or aluminum shall not be used.

3.2.17.2 Identification plates shall be secured to equipment with corrosion-resistant metallic screws or rivets.

3.2.17.3 Pump identification plates shall contain data as follows:

(a) Manufacturer’s name.
(b) Manufacturer's model or type and size.
(c) Service application.
(d) Manufacturer’s serial number.

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(e) Salient design characteristics:
(1) Capacity in gallons per minute (gal/rein).
(2) Total head in pounds per square inch (lb/in^2).
(3) Speed of shaft in r/rein.
(4) Brake hp.
(5) Test pressure (omit for submarine seawater pump).
(6) Special data vital to the unit:
a, Suction pressure (omit for submarine seawater pump).
b. Submergence (omit for submarine seawater pump).
c. Impeller diameter.
(f) Contract number and item number for multiple unit orders.
(g) National stock number.
(h) Section for Defense Contract Administration Services Management Area (DCASMA) stamp.
(i) NAVSEA technical manual number.
(j) Certification data drawing number.

3.2.17.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.2.18 Piping, valves, and threaded items.

3.2.18.1 Valves, flanges, fittings, and bolting for pipe connections shall conform to MIL-STD-438 for submarines and MIL-sTD-777 for surface ships and as specified in the contract or order (see 6.2.1). All pump flanges, except for welded connections, shall have a face circular lay finish as detailed in the applicable general note of MIL-STD-438 for submarines and MIL-STD-777 for surface ships.

3.2.18.2 Tapered pipe threads are acceptable for use in grease fittings located on nonmoving parts. Fittings not covered by MIL-STD-438 and MIL-STD-777 for application when ship connections are not involved shall be submitted for review by the design review agency.

3.2.18.3 Threaded plugs, threaded connections, and threaded closures are prohibited in the pressure boundary of submarine pumps.

3.2.19 Locking devices. Internal threaded fasteners and threaded machine elements shall be secured by locking devices where continuous operation under the condition of shock, vibration, and temperature specified herein depend on maintaining tight connections of parts. All setscrews internal to the pressure boundary of the pump, and all setscrews on internal and external rotating portions of the pump shall be secured by locking devices. Locking devices shall be subject to review by the design review agency.

3.2.20 Reverse rotation. Motor driven pumps shall withstand reverse rotation for a period of 1 minute at all speeds up to design speed without damage.

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3.2.21 Provision for handling. Eyebolts, lugs, holes, and other means shall be provided to permit attachment of lifting gear for lifting the assembled pump, driver, and attached accessories as a complete unit. Means shall be provided for the handling of parts and components weighing 150 pounds and over which cannot be handled manually during unit overhauls and preventive maintenance inspections.

3.3 Detail requirements.

3.3.1 Unless otherwise specified (see 6.2.1), horizontal main circulating pumps shall be flexible coupled, and vertical main circulating pumps shall be rigid coupled.

3.3.2 Performance characteristics.

3.3.2.1 The performance characteristics of the pump shall be as specified (see 6.2.1).

3.3.2.2 Main circulating pump suction conditions shall be as specified in 6.2.1(d)(6).

3.3.2.3 Unless otherwise specified (see 6.2.1), pumps for surface ships shall be provided with bilge connections, which shall be capable of pumping bilges at a dynamic suction lift of 15 feet of water.

3.3.2.4 Unless otherwise specified (see 6.2.1), pumps for submarines shall not have a bilge connection.

3.3.3 Casings.

3.3.3.1 The pump casing shall be split to facilitate the ready removal and replacement of pump shaft, internal pump bearings, and impellers. The casing of all vertical pumps, except those for submarine application, shall be split vertically in such a manner as to allow the removal of the rotor without disturbing the piping attached to the pump casing or to the driving unit.

3.3.3.2 Unless otherwise specified (see 6.2.1), the bilge suction connection shall be provided on a fitting adjacent to the pump inlet by the pump manufacturer.

3.3.3.3 Casings shall be sufficiently rugged to withstand without fracture or appreciable distortion the strains to which they may be subjected. The nozzle loadings to which the pump will be subjected will be calculated by the shipbuilder and forwarded to the pump contractor by the contracting activity (see 6.2.1). Under the combined effect of the nozzle loads and design pressure, no parts of the casing or other pump parts shall exceed yield strength and no rotating clearances shall make contact.

3.3.3.4 Clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools. Ordinary tools are tools which are listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the DCASMA).

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3.3.3.5 Fitted bolts or dowel pins shall be provided to insure maintenance of alinement in assembly. Dowel pins shall be corrosion-resistant, shall have a threaded end and nut design, and shall be secured against coming adrift under shock loading.

3.3.3.6 Forcing bolts shall be provided for breaking joints.

3.3.3.7 Casings of horizontal and vertical surface ship pumps shall be fitted with drain and vent connections. Drain connections shall be in accordance with MS16142 and MS18229 and shall permit complete drainage of the pump without disassembly of the pump. For vertical, bottom suction pumps, the pump suction may serve as the drain. Threaded drain connections are prohibited in the pressure boundary of submarine pumps.

3.3.3.8 For type I pump, the casing shall be fitted with a removable casing liner in way of propeller.

3.3.3.9 For type II and type III pumps, the casings shall be fitted with removable casing wearing rings. The casing wearing rings shall be located and shaped in such a manner that leakage through wearing ring clearances shall not be allowed to impinge directly on the casing.

3.3.3.9.1 Spare casing wearing rings (see table 11) shall be machined undersize by at least 0.050 inch on the diameter in order to permit finish machining during overhaul installation to mate with machined-down used impellers.

3.3.3.10 Pump casing joints shall be made up using compressed asbestos sheet gaskets. O-rings may be used to seal casing joints in the sea water pumps for submarines. O-rings and gaskets are prohibited from being in contact with highly-alloyed corrosion resistant steel parts.

3.3.3.11 Venting. Pump casing shall be self-venting or be provided with casing vents. Those portions of the casings which might trap air, but are not vented, when the pump is secured, shall be rapidly cleared of air by flow during pump operation. Special attention shall be paid to the vent location so that the seal lubrication requirements of 3.3.8.5 are met during pump start-up and operation.

3.3.3.12 For pumps with suction and discharge connections which are prepared for welding to the system piping, the vent connections shall be located on parts of the pump which do not need to be removed for repair or maintenance and shall be configured for welding the system venting piping directly to the pump. If, in order to perform its intended function, the vent connection must be located on a removable part of the pump, then the vent connection shall be flanged for mating with flanges.

3.3.3.13 Casing thickness shall include an allowance for fabrication deviations and for erosion and corrosion in service. For submarine pumps, areas subject to fluid velocity in excess of 15 feet per second at rated flow shall be readily renewable or shall be protected by replaceable erosion shields.

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3.3.4 Suction and discharge connections.

3.3.4.1 Unless otherwise specified (see 6.2.1), suction and discharge connections shall be flanged for mating with flanges in accordance with MIL-STD-777 or MIL-STD-438 for the applicable system.

3.3.4.2 Suction and discharge connection of split case pumps shall be on the fixed half of casing,

3.3.5 Impellers or propellers.

3.3.5.1 Outside surfaces of impellers shall be smooth finished.

3.3.5.2 Impellers shall be keyed on the shaft and securely held against movement by locked nuts or other means.

3.3.5.3 Impellers, for type II and type III pumps, shall not be fitted with wearing rings. The impeller hub wearing surfaces shall have material thickness to permit reducing the diameter of the impeller hubs by as much as 0.050 inch to accommodate undersize casing rings to restore design running clearances.

3.3.5.4 Each impeller shall be dynamically balanced in accordance with MIL-STD-167-1 .

3.3.5.5 O-rings and gaskets are prohibited from being in contact with highly-alloyed corrosion resistant steel parts.

3.3.6 Shafts.

3.3.6.1 Shaft threading, except threading for rolling contact bearing lock nuts, shall be counter to the direction of rotation. Shaft threading for rolling contact bearing lock nuts may be clockwise or counterclockwise.

3.3.6.2 The shaft (pump or driver) except those driven by a.c. motors, shall be provided with means to permit the ready and direct use of portable tachometers. When this is impracticable, provision shall be made for the use of a portable tachometer on the shaft of the driver.

3.3.6.3 Shafts shall be fitted with shaft sleeves in way of all wearing surfaces, such as stuffing boxes, bushings, and water lubricated bearings. O-rings shall be installed between the shaft sleeve and shaft.

3.3.6.4 Shaft flingers shall be provided adjacent to the gland for all pumps. The flinger shall be so located that leakage which might occur between the gland shaft sleeve and the shaft, as well as from the gland, is not allowed to reach pump or motor ball bearings. For rigid coupled units, the coupling may serve as the pump flinger provided the coupling is constructed of corrosion-resistant material.

3.3.6.5 A balance ring shall be located in the coupling area between the lower motor bearing and the pump stuffing box, allowing sufficient clearance for packing gland adjustment. The balancing ring shall include the features and ounce-inch capacity specified in MIL-E-917 for submarine rotating equipment balance rings.

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3.3.7 Rotating assembly.

3.3.7.1 Rotors shall be dynamically balanced with all rotating parts connected thereto; for common shaft assemblies, or those using rigid coupling, this requires dynamic balance with the rotating element of the driving unit in place. However, rotating parts may be balanced individually provided that, when assembled, the imbalance shall not exceed the limits specified in MIL-STD-167-1. .

3.3.7.2 Where balancing is required as a maintenance procedure to maintain proper vibration or noise performance of the pump and driver unit in service, the pump shall be designed such that balancing may readily be performed by overhaul activity personnel.

3.3.7.3 The pump design shall be such and calculations shall be made to demonstrate that the maximum operating speed of the pump will be not greater than 70 percent of the first critical speed.

3.3.7.4 Impellers, shafts, and other parts of the rotating assembly shall not contain radially oriented setscrews except setscrews solely used as balancing weights in the balancing ring design required by 3.3.6.5 are acceptable.

3.3.8 Stuffing boxes and mechanical shaft seals.

3.3.8.1 Pumps shall be provided with mechanical shaft seals. The stuffing box for surface ship pumps shall accommodate a minimum of two rings of packing for use in the event of a mechanical seal failure. The mechanical shaft seal shall have a tungsten carbide against a carbon sealing surface. Mechanical seal housing material shall be as specified in table I.

3.3.8.2 Sufficient space shall be provided between bearings and stuffing boxes to permit easy inspection of bearings and mechanical seal leakage.

3.3.8.3 Pumps for submarine application shall be provided with pressure breakdown device. The pressure breakdown device shall be located in way of a shaft sleeve and not in way of the impeller. The device shall limit the leakage to 5 gal/rein in the event of failure of the mechanical seal, under conditions of maximum suction pressure specified. The stuffing box for submarine pumps shall accommodate a minimum of four rings of packing for use in the event of a mechanical seal failure.

3.3.8.3.1 Pumps shall be configured so that normal gland leakage, whether from mechanical seal or when operating under emergency packing, shall be collected and piped to waste. The collection area shall be tapped for drain connections so that the leakage can be drained away from the pump casing foundation, bearing housings, or driving units. With the pump running or shut down, the collection area shall collect and drain all gland leakage, up to the greater of either (a) one quart per minute or, (b) three times the minimum allowable leakage rate recommended by the contractor and approved by the design review agency for normal operation under packing.

3.3.8.4 Mechanical shaft seals shall be positioned on the shaft by means of stub or step sleeves. Mechanical shaft seals shall not be positioned by use of setscrews alone.

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3.3.8.5 The design shall insure that positive liquid pressure is supplied to the seal faces under all conditions of operation and that there is adequate circulation of the liquid at the seal faces to minimize the deposit of foreign matter on the seal parts. The pump certification data shall contain certification by the seal manufacturer that there will be adequate circulation of liquid and adequate lubrication at the seal faces when the seal is installed as shown on the drawing.

3.3.8.5.1 Submarine pumps shall be fitted with mechanical seals and cyclone separators. Surface ship pumps shall be fitted with cyclone separators when the pump discharge pressure is 30 lb/in^2 gage or above.

3.3.8.6 Stuffing box glands shall be set up by nuts threaded on studs secured in the castings.

3.3.9 Coupling.

3.3.9.1 For horizontal units, an all metal flexible coupling shall be installed between the pump and driving unit. Flexible couplings shall be in accordance with type II, class 2 of MIL-C-23233.

3.3.9.2 Vertical units shall be fitted with a rigid spacer-type coupling between the pump and driving unit. The design of the pump and coupling shall permit mechanical seal replacement by removing the spacer without unbolting, lifting, or otherwise disturbing the driver, the pump cover, or the pump casing.

3.3.9.3 Flexible coupling hubs shall be keyed to the shafts and secured by lock nuts. The hubs shall be fitted on a taper with keys parallel to the taper.

3.3.9.4 Rigid couplings shall have fitted bolts and the coupling flanges shall be marked at assembly to insure proper reassembly after overhaul of the unit.

3.3.9.5 Horizontal flexibly coupled units with bedplates shall have coupling guards provided. Guards shall permit ready access to the coupling for lubrication and examination.

3.3.10 Material. The materials of the pump shall conform to the materials specified in table I. However, this specification is not intended to be restrictive provided proposed alternative materials will give equal or better service than the material specified. Proposed alternative materials shall be subject to approval by the design review agency. Components of the pump for which the specific materials are not specified shall be materials best suited for the intended service. Materials which can be sensitized and are subjected to heat treatment in the sensitization range during fabrication shall be capable of passing the intergranular corrosion tests of the base material specifications. Particular attention shall be given to avoiding sensitization of materials during hard facing, stress-relieving, or repair welding.

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TABLE I. Materials.

Application[1/2/11/12/]	Material	Specification	ASTM publication
Casings (all types) and diffusers for type II and III pumps:			
Surface ships	Copper-nickel (70-30)	MIL-C-20159	B 369 UNS No. C96400
Submarines	Wrought copper-nickel (70-30)	MIL-C-15726 (hot forged)[3/]	----------
	Copper-nickel (70-30) permitted only for diffusers not contributing to pressure boundary strength	MIL-C-20159	B 369 UNS No. C96400
Casing liners (type I pump)	Copper-nickel (70-30)	MIL-C-20159	B 369 UNS No. C96400
Diffusers (type I pump)	Copper-nickel (70-30)	MIL-C-20159	B 369 UNS No. C96400
	Copper-nickel	MIL-C-15726 (hot forged)	----------
Casing bolts, studs and nuts[4/5/7/]	Ni-Cu-Al alloy	MS18116	----------
	Nickel-copper alloy	MIL-N-24106, class C	
		QQ-N-281, class A	B 164, class A or B
Impellers or propellers	Highly-alloyed corrosion resisting steel	----------	A 744, CN-7M or CN-7MS[6/]
Impeller nut[5/7/]	Nickel-copper alloy	QQ-N-281, class A	B 164, class A
	Ni-Cu-Al alloy	QQ-N-286	----------
Impeller nut washer	Wrought highly-alloyed corrosion-resisting stainless steel	----------	B 463 (plate) B 473 (flat bar) UNS N08020
Impeller Nut locking device	Ni-Cr-Mo-Cp alloy	----------	B 446, B 443 UNS N06625

See footnotes at end of table.

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TABLE I. Materials. - Continued

Application[1/2/11/12/]	Material	Specification	ASTM publication
Shims, spacer, metallic mechanical seal parts (except springs) and other small parts not listed in this table which are in close proximity to impeller	Wrought highly-alloyed corrosion-resisting stainless steel Ni-Cr-Mo-Cb alloy Highly-alloyed corrosion resisting steel	---------- ---------- ----------	B 463 (plate) B 473 (bar) UNS N08020 B 446, B 443 UNS N06625 A 744, CN-7M or CN-7MS
Shims, spacers metallic mechanical seal parts (except springs) and other small parts not listed in this table which are not in close proximity to impeller	Nickel-copper alloy Ni-Cu-Al alloy Ni-Cr-Mo-Cb alloy	QQ-N-281, class A QQ-N-286[7/] ----------	B 164, class A ---------- B 446, B 443 UNS N06625
Mechanical seal springs	Ni-Cr-Mo-Cb alloy	----------	B 446, UNS N06625
Casing wearing rings	Ni-Cu-Si alloy centrifugally cast	MIL-C-15345, alloy 16	----------
Shafts[5/7/]	Ni-Cu-Al alloy Nickel-copper alloy	QQ-N-286, class A QQ-N-281, class A	---------- B 164, class A
Journal sleeves	Ni-Cu-Al alloy (Brinell hardness 265 minimum)	QQ-N-286, class A	----------
Shaft sleeves[7/8/9/]	Ni-Cu-Al alloy (Brinell hardness 165 minimum) Nickel-copper alloy	QQ-N-286, class A QQ-N-281, class A	---------- B 164, class A
Bearings (internal water lubricated) [10/]	Bearing bronze	QQ-C-390, alloy 910 or 934	----------

See footnotes at end of table.

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TABLE I. Materials. - Continued

Application 1/ 2/ 11/ 12/	Material	Specification	ASTM publication
Pump internal locking components 7/	Nickel-copper alloy Ni-Cu-Al alloy Ni-Cr-Mo-Cb alloy	QQ-N-281, class A QQ-N-286	B 164, class A --- B 446, B 443 UNS N06625
Cyclone separators and piping	Nickel-copper alloy 13/ Copper-nickel (70-30) 13/	QQ-N-281, class A MIL-C-15726	B 164, class A ---
Labryinth seal floating ring	As specified (see 6.2.1)		
Labryinth seal stationary ring	As specified (see 6.2.1)		
Labryinth coil spring	Ni-Cr-Mo-Cb alloy	---	B 446, UNS N06625
Labryinth seal lock pin	Ni-Cr-Mo-Cb alloy	---	B 446, UNS N06625 B 443
Bolts, studs, and nuts (pump to support bracket)	Steel	MIL-S-1222	A 193, grade B7 A 194, grade 2H
Bed plates	Cast steel or structural steel	MIL-S-15083, grade CW	---

1/ Hardware or subassembly parts which are used in a major pump part listed in table I and which perform the same or similar functions as the major part (e.g., casing drain plug) shall be of a material specified for the major part.

2/ Detail drawings and lists of material shall reference the specifications actually followed in each case, and shall include the class, type, or grade of material used in each case, as applicable (see 3.3 and 3.6.2.1). ASTM materials shall not be used for submarine submergence boundary parts.

3/ Yield strength at 0.5 percent extension under 18,000 lb/in^2 minimum, 48,000 lb/in^2 maximum.

4/ Design of bolts, studs, and nuts shall be in accordance with MIL-S-1222. Nickel-copper-aluminum threaded fasteners shall be as specified in MS18116.

5/ Nickel-copper alloy shall not be used for both male and female threaded members in the same assembly. If bolts and studs in accordance with QQ-N-281, class A, are used, they shall be hot finished for submarine application. Any nickel-copper bolts and studs made of cold drawn, stress-relieved material for use in submarine pump parts, shall be in accordance with MIL-N-24106, class C in lieu of QQ-N-281.

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6/ A certificate of compliance with ASTM A 744 shall be provided. In addition, certification of the chemical test results for each heat of material and the specific heat treatment (including temperature and holding time) to which each casting was subjected shall be provided. The time between removing the stainless steel part from the furnace and quenching shall be minimized. This certification shall list both chemical test result values and the corresponding acceptance criteria valve specified in ASTM A 744. Repair welding shall be accompanied by post-weld heat treatment in accordance-with ASTM A 744 (see 6.2.2).

7/ All nickel-copper-aluminum alloy parts shall be annealed prior to age-hardening except parts for which a hardness greater than 265 BHN is required.

8/ For submarine pumps, sleeves which are a seat for O-rings where movement occurs, e.g., under mechanical seals which are spring loaded, shall be nickel-copper alloy that is weld clad with Ni-Cr-Mo-Cb alloy. The area to be weld clad shall be minimized. Cladding shall extend a minimum of 3/16 inch deep in the final machined condition. Deposit shall be made by a minimum of two passes. The filler weld material shall be as follows: Covered electrode in accordance with MIL-E-22200/3 (MIL-1N12), or bare wire in accordance with MIL-E-21562 (MIL-EN625). Machining into the first (lower) layer for the final surface shall not be permitted. The final machined surface of the cladding shall be PT inspected. The acceptance standard shall be class I of NAVSEA 0900-LP-O03-8000. The surrounding areas shall be inspected in accordance with MIL-STD-278.

9/ Sleeves that are subject to wear or that establish a critical clearance (i.e., less than 40 roils radial clearance) shall be made of nickel-copper-aluminum alloy and shall meet the 265 Brinell minimum hardness specified in the table. Sleeves for which footnote 8 requires weld cladding shall be made of nickel-copper alloy. Other sleeves shall be made of either: (a) nickel-copper-aluminum alloy meeting the annealing requirement of footnote 7 rather than the above hardness requirement or, (b) nickel-copper alloy.

10/ Special high lead bronze or other suitable bearing materials may be used subject to review by the design agency. Bearing metal, alloy B-10, is an acceptable alternative material.

11/ When the mechanical properties of the material used in the fabrication of a part are altered by metal working processes or heat treatment, sample material from each lot of material, as defined in the material specification, shall be subjected to tensile tests in the final heat treated condition for the part. As a minimum, tensile tests shall determine ultimate strength, yield strength, percent reduction in area, percent elongation and hardness. The tensile tests shall be conducted on test samples machined from the heat treated material or from a piece of parent stock which had been heat treated in the same batch in the lot processed at the same time. The tensile test requirements and acceptance criteria for mechanical properties shall be included on the part drawing. Acceptance criteria for the mechanical properties shall be in accordance with the material specification or as approved by the design review agency. In no case shall ductility of less than 10 percent elongation in 2 inches be allowed.

12/ Where nickel-aluminum-bronze is used, it will be in accordance with alloy 632M of QQ-C-465 and its amendment.

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13/ Cyclone separator parts exposed to water velocities greater than 15 feet per second shall be made of nickel-copper alloy. Other cyclone separator parts shall be 70-30 copper-nickel except: (a) parts such as fittings, not readily available in 70-30 copper-nickel, may be made of nickel-copper alloy and (b) parts that would be unacceptably galvanically attacked if made of 70-30 copper-nickel shall be made of nickel-copper alloy.

3.3.10.1 Recovered materials. Unless otherwise specified herein, all equipment, materials, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Tools. Tools which are available in the Federal Supply Catalog, such as common wrenches and standard pullers and extractors, will not be required. Standard complete or limited sets of pullers or extractors or parts of puller sets in accordance with GGG-P-781 or GGG-E-950 shall be identified as to type and use in notes on the outline or sectional assembly drawings. Special tools shall be detailed and included in the list of material. Special tools required for the maintenance and repair of the pump units shall be furnished. The number of sets of special tools of the pump units shall be as specified (see 6.2,1). Special wrenches shall be of forged steel with hardened jaws. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of the catalog may be consulted in the office of the DCASMA).

3.5 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.6 through 3.10.2.

3.6 Drawings. Unless otherwise specified (see 6.2.1), the contractor shall prepare drawings in accordance with the data ordering document included in the contract or order (see 6.2.2) and the unique features specified in 3.6.1 through 3.6.4 shall be included.

3.6.1 Drawing content.

3.6.1.1 General requirements.

3.6.1.1.1 "Manufacturer's Use Only” notes. Information intended for manufacturer's use only shall be so designated. Conversely, on a commercial drawing with wide usage it is permissible to designate portions thereon, “For Navy use only.”

3.6.1.1.2 Dimensional tolerance. Unless otherwise indicated by note, legend, or key all tolerances shown on a drawing shall be complied with during both manufacture and maintenance repair.

3.6.1.2 Design drawings. Design drawings shall consist of the drawings described in 3.6.1.2.1 through 3.6.1.2.4.

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3.6.1.2.1 Sectional assembly drawings. Sectional assembly drawings shall include a sectional assembly with references to the list of material for identification of individual parts. All running clearances shall be shown and shall be dimensioned and labled as diametral clearances. (Diametral clearance equals the difference between the diameters of mating parts.) This may necessitate some side or partial view in order to show parts not otherwise shown in the main section view. Tightening torques with tolerances and thread lubrication requirements for threaded fasteners shall be shown on the assembly drawing.

3.6.1.2.2 List of material. The list of material shall include every part required in the pump assembly, including those parts not required to be detailed. The list of material shall include an indication of each part required by this specification to be furnished as an on board repair part (this is not a repair parts list). The list of material shall include the manufacturer's drawing number and service part number for all parts for which detail drawings are required (see 3.6.1.2.3) as well as quantity required, material type, and material specification for all parts. Parts shall be named to indicate the function they serve. Parts for which detail drawings are not required shall be identified as to dimensions. The list of material may be shown on the sectional assembly drawing.

3.6.1.2.3 Detail drawings. Detail drawings shall be furnished of all parts and sub-assemblies necessary for evaluation of the equipment, and of all parts necessary for maintenance, and overhaul of the pumps. Details of these parts shall be-so complete as to permit emergency manufacture by a Naval shipyard without assistance from the original manufacturer. Details of pump casing shall be furnished, but the dimensioned development of cored hydraulic passages need not be shown. However, for all cast parts that carry substantial loads under shock, the detail drawings shall specify minimum cross-sectional areas (whichever is more appropriate), Sub-assemblies whose parts cannot be acquired or serviced individually, shall be shown as a single part and so indicated. Multi-detail drawings are preferred, but monodetail drawings may be used.

3.6.1 .2.3.1 Impeller machining and inspection drawing. Detail drawings shall explicitly require that each impeller, including spare and replacement impellers, be dynamically balanced to less than the maximum ounce-inch residual imbalance allowed by MIL-STD-167-1 (type II) for the impeller's weight and rotational speed. For submarine pumps, the impeller detail drawing shall: (a) show the dimensions and shape of the discharge vane tip, including any under-filling or over-filling details, (b) specify sufficient dimensions and tolerance, to be recommended and justified by the manufacturer, to prevent significant impeller-to-impeller variation in required net positive suction head due to manufacturing variations including casting variations. Significant variation is defined as that which would be large enough to cause a spare or replacement impeller to not pass the net positive suction head test of 4.6.3(b) if such impeller were to be tested and (c) explicitly require that each impeller, including spare and replacement impellers, be inspected and comply with the above dimensions and tolerances.

3.6.1.2.3.2 Drawings showing the application of studs requiring the use of a locking resin shall have a notation requiring: (a) manufacturer's recommendation regarding the use of primer, (b) manufacturer's recommended primer, and (c) the minimum and maximum waiting times associated with the use of the primer and resin. The notation shall require strict adherence to these requirements.

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3.6.1.2.3.3 Drawings that include springs, such as for some mechanical seals and labyrinth seals, shall:

(a) State the number of coils, the coil diameter, the wire diameter, the material, the free length, the solid length, the nominal working length, the working length force and the plus and minus tolerance on that force.
(b) Include a requirement that quality control testing be done on production springs to ensure the springs return to their original length after being fully compressed and to ensure the force at working length is within tolerance. The drawing shall identify the sampling plan and the acceptance criteria.

3.6.1.2.4 Outline drawings and certification data.

3.6.1.2.4.1 Outline drawings and certification data shall be in the form of supplemental drawings to all sectional assemblies and detail. Separate outline drawings, drawing lists, and certification data shall be furnished under each contract or order unless the complete equipments covered by the outline drawing and the referenced drawings are in fact identical in all respects.

3.6.1.2.4.2 The outline drawings in addition to the certification data required by DOD-D-1000 shall contain the following:

(a) Dimensional outline assembly drawing of the pump with its prime mover, bedplate and attached auxiliaries.
(b) Complete performance data of pump, prime mover, and attached auxiliaries , if applicable.
(c) Complete equipment performance curves, based on actual tests (the original submission may show design performance curves vice test curves). The serial number of the unit whose tests results are shown shall be written on the curve, unless the curve is predicted, in which case “Predicted” shall be written. The actual test data shall be that of the first performance tested unit, which 4.6.3 requires be used as the head repeatability acceptance criteria basis for the following units.
(d) Weights of major sub-assemblies over 50 pounds, e.g., rotor, upper half-casing, impeller and the weight of the complete unit including driver, dry and in operating condition.
(e) Shipbuilder’s connections showing size, type and dimensions of flanges.
(f) Center of gravity of the pump component and of the complete assembly.
(g) Radii of gyration of complete assembly about each of the three principal axes.
(h) Identification of system in which installed.
(i) Critical speed.
(j) Maximum thrust loads.
(k) Endurance tests, high-impact shock tests and nondestructive tests (if performed) and the design review agency letters approving these reports or extension of any tests.

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(l) Overall dimensions of the complete unit including locating of all fittings and connections, the space required for removal and replacement of parts for maintenance, and mounting information.

(m) The certification by the mechanical seal manufacturer required by 3.3.8.5.

(n) The following dimensions, which are related to interchangeability of pumps and drivers as a unit. The drawing shall include a tolerance for each of these dimensions. Each dimension shall also be shown with the same value and tolerance on the detail drawing for the part:

(1) For support feet or other support connections between the ship and the pump and driver unit: The X, Y, and Z dimensions, or three dimensions in another suitable coordinate system, locating the feet and any bolt holes in the feet.

(2) For all piping connections: The X, Y, and Z dimensions, or three dimensions in another suitable coordinate system, locating the point of intersection between the axis of the connection and the face of connection, such as a flange face. The dimensions shall be referenced to the same surfaces to which the support feet are referenced, either directly or via other tolerance dimensions.

(3) For all flanged piping connections:

a. The diameter of the bolt circle and the concentricity of that circle relative to the axis of the connection.

b. The angular location of each bolt hole with respect to some reference line shown on the drawing. Even if the angle to the reference line is 0 to 90 degrees, that number shall be stated, with a tolerance. It is acceptable to locate only one bolt hole relative to the reference line and to state the remaining holes are equally spaced as long as a tolerance is specified on the equalness of the spacing.

c. In lieu of a. and b. above, an X-Y coordinate system may be used to locate the bolt holes.

d. The diameter of the bolt holes.

e. The tolerance on perpendicularity between the flange face and the axis of the connection.

f. The inner diameter of the suction and discharge nozzle at the flange face and the concentricity of that diameter relative to the axis of the connection.

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3.6.1.2.5 On board repair parts list. A list of on board repair parts shall accompany the outline drawings. This list shall be prepared on provisioning list forms in accordance with MIL-P-15137. This list shall be furnished whether or not the parts are required in the pump contract or order. This list shall not be modified to indicate parts or quantities of parts furnished under supplementary or separate contracts or orders. The list is intended as a record of recommendations of the manufacturer and purchaser at the time of equipment drawing approval. The list of on board repair parts actually purchased shall be prepared as required by MIL-P-15137 (see 6.5.2).

3.6.1.2.6 Review. Design drawings shall be submitted to the design review agency for approval (see 6.2.1).

3.6.1.3 Basic design drawings. Basic design drawings covering pumps under this specification are intended for submittal to the design review agency prior to and independent of invitations for bids. As such these are different from design drawings as described in 3.6.1.2. Each drawing may be a range of sizes of pumps so long as they are of the same basic design.

3.6.1.3.1 Contents of basic design drawings. These drawings shall consist of small. scale (not necessarily to scale) plan and elevation views showing overall dimensions. If the plans cover a family of pump sizes all of the same design, the dimensions may be tabulated. The range of capacities, pressure and speeds shall be shown. The main part of this drawing shall be an undimensioned sectional assembly with complete list of material which shall include every part required in the pump assembly. If materials of some parts may change with various pump characteristics or with differing fluids to be pumped, a table shall be included to show these variations. Alternate subassembly arrangements of different parts may be included as desired.

3.6.1.3.2 Detail drawings of individual parts are not required in connection with basic design drawings, but may be included if desired by the manufacturer for clarification, or to obtain advance approval of the design review agency.

3.6.1.4 On one of the design delivered drawings described in 3.6.1.2, the mechanical seal contractor shall identify and certify the following:

(a) Whether or not there is an adhesive or other setting compound factory-installed on the elastomer bellows or O-ring that seals between the mechanical seal and the impeller hub or shaft sleeve.
(b) The description of any such adhesive or setting compound.
(c) Any time limit associated with that compound when installed on a mechanical seal in stowage and any time limit between initial wetting of that compound and the final positioning of the mechanical seal in the pump.
(d) The type and amount of lubricant, if any, to be used to facilitate axial positioning of the mechanical seal in the pump during seal installation. The lubricant shall be silicone compound in accordance with MIL-S-8660, and all elastomers, O-rings, or seals shall be compatible with the silicone compound. The drawing shall require the manufacturer to use that lubricant during pump assembly.

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3.6.2 Material identification.

3.6.2.1 Perferred material reference. Where materials of identical or equal quality can be more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced the following is the order of preference:

(a) Federal specification or standard.
(b) Military specification or standard.
(c) Industry and Technical Society specification or standard.
(d) Manufacturer's specification or standard.

3.6.2.2 Material substitutions. Where materials other than covered by (a), (b), and (c) of 3.6.2.1 are referenced and approved by the design review agency, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the material in terms of the nearest Federal or Military specification, in order to enable Naval repair facilities to make emergency repairs as necessary from Navy material stocks.

3.6.3 Drawing identification. Prime contractors who purchase items from subcontractors shall use the subcontractor’s drawing number as the single reference identification in all cases where the part(s) delineated thereon are produced by the subcontractor. The prime contractor shall not add his drawing number to the drawing except as an unofficial reference outside the drawing border or margin.

3.6.3.1 Prime contractors who purchase semi-finished parts from subcontractors for final production, test or selection phases in their own shops, have the option of using as the single drawing identification either their own title block and drawing number or the title block and drawing number of the subcontractor, but not both.

3.6.4 Drawings of drivers and associated equipment shall be in accordance with the specifications covering those equipments,

3.7 Technical manuals.

3.7.1 Unless otherwise specified in the contract or order (see 6.2.1), technical manuals shall be furnished in accordance with the data ordering document included in the contract or order (see 6.2.2). The modifications and clarifications specified in 3.7.2 through 3.7.5 shall apply. A separate manual shall be furnished for each different pump unit except in special cases when manuals covering more than one pump may be approved. Preliminary manuals shall be submitted to the design review agency for approval and shall include all proposed sections complete. Unless otherwise specified in the contract or order (see 6.2.1), manuals shall include pump, driver, controls and all appurtenances and all applicable drawings. Performance curves for final manuals may be furnished as insert pages after delivery of hardware.

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3.7.2 Manuals covering pumps driven by turbines or engines shall include drawings and full operation and maintenance instructions covering the turbines, engines, governors and any other applicable accessories.

3.7.3 Manuals covering pumps driven by conventional electric motors with conventional controllers need not include operation and maintenance instructions covering the motors and controllers. They shall, however, include master drawings and certification data covering the motors and controllers. Complete operation and maintenance instructions covering electrical equipment shall be included when specifically required (see 6.2.1).

3.7.4 Each manual shall include not less than the following illustrations covering the pump and driver:

(a) Sectional assembly drawing.
(b) Outline drawing.
(c) Complete list of material corresponding to the sectional assembly drawing.
(d) Certification data,
(e) A minimum of two photographs of the complete unit, pump and driver, taken 180 degrees apart on a horizontal plane. Where several sizes of identical design units are being supplied, photographs of one design unit only need be supplied.

3.7.5 The alinement procedure instructions in the manual shall describe in complete detail the means by which the required alinement clearance at sleeve bearings and wearing rings is to be established. The manual shall prescribe the maximum permissible eccentricities which can be tolerated in the aforementioned areas with the pump in the assembled condition. Exceptions to any part of the aforementioned requirements in any specific area may be had only by providing for the design review agency approval a written technical justification for the said exception. The manual shall specify the maximum allowable clearance for wearing rings and sleeve bearings before replacement of the part(s). In addition, the manuals shall contain:

(a) Lubrication schedule including lubricant and procedure.
(b) A statement directing the user of the technical manual to see the Centrifugal Pump Chapter of the Naval Ships Technical Manual for the criteria for replacing mechanical seals due to leakage. The pump technical manual shall not redundantly specify the same or different criteria.
(c) Torque values and sequence required to maintain proper seal and alinement.
(d) In-field dynamic balancing instructions.
(e) The mechanical seal installation procedure including complete information regarding adhesive, the adhesive time limits and lubricant (see 3.6.1.4), and any additional guidance needed to avoid causing an excessive preload on the mechanical seal.

3.7.6 The quantity and distribution of manuals shall be as specified (see 6.2.1).

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3.8 Design data and calculations. Design data and calculations and design records shall be submitted for approval of the design review agency for each pump class, design and size. Submittal shall be not later than submittal of design drawings for approval. It is intended that the design data and calculations shall be in sufficient detail to describe the design, physical appearance and internal assembly to permit release for manufacture References listing the author, publication, volume or text for all assumptions made in the calculations and for all formulae or methods of computation not readily identifiable shall be cited.

3.8.1 Design performance curves. The following curves shall be provided:

(a) Head versus flow.
(b) Hydraulic efficiency versus flow.
(c) Net positive suction head required versus flow.
(d) Brake hp versus flow. This curve shall demonstrate that the motor is not overloaded under any condition of suction head, discharge head or free delivery possible within the pump application

3.8.2 Design features. A discussion of the following design features shall be provided:

(a) Number of stages and reasons for selection.
(b) Method of shaft sealing.
(c) Type of coupling including previous service experience.
(d) Those areas in which system design conditions, as furnished by the design review agency, adversely affect pump reliability or result in abnormal design features.
(e) Steps required for disassembly and reassembly (including alinement) to perform routine maintenance.
(f) For sea water pumps the galvanic, wear, and galling compatibility of the materials used for parts in contact with the pumped fluid.
(g) Any other design features which the manufacturer considers of significant interest to describe the proposed unit.
(h) Whether the specified rated capacity and range of capacity (see 6.2.1) will result in shipboard operation at capacities that will cause pump internal recirculation (suction or discharge) sufficient to degrade pump life, performance or reliability.

3.8.3 Use of materials other than as specified herein shall be justified. It is not intended that materials which conform to and are used as specified herein shall require justification.

3.8.4 Calculations shall be submitted to cover the following:

(a) Shafting analysis of pump and driver assembly including calculation of critical speed. If calculations are done by a computer, the method of analysis shall be described and only the results shall be submitted with identification of corresponding pump parameters used in the calculations. The following shall form the basis of the critical speed calculation:

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(1) The effects of all major changes in shaft cross section moment of intertia shall be included.
(2) No support for the shaft shall be assumed other than at the designated bearings.
(3) All masses in the rotating system, including the effect of entrained fluids, shall be lumped at discrete points along the shaft.
(4) No support for any shafting shall be assumed to be provided by additional shafting when two sections of shafting are separated by a flexible coupling, nor should any flexibility in bending or shear be assumed for a rigid coupling.
(5) For pumps which operate under variable speed conditions, the operating speed shall be construed to be the maximum operating speed.

(b) Loads imposed on the shaft and bearings. The calculations shall include the following:

(1) Axial and radial loads and load directions. Loading shall be calculated for the design point and also for the highest loaded conditions over the operating range (see 6.2.1). These loads shall include all operating loads together with those that might be induced by thermal expansion, by bearing preload, and by build-up of tolerances in the completed assembly and by ship roll, pitch, list, and trim. The radial loads at the impeller shall be calculated by the following:

$$P = \frac{KHD_2B_2}{2.31}$$

Where:

P = the radial load in pounds.

H = head in feet.

D_2 = impeller outside diameter, inches.

B_2 = impeller overall width including shroud, inches.

K = a constant which varies with capacity.

Unless data, supported by tests, justifies the selection of a lower value, K shall be in accordance with the following. Lower K values may be used only if specifically approved by the design review agency.

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Casing type	At shutoff	At rated flow	At 140 percent rated flow
Single volute	0.00024Ns for Ns less than 1500 0.36 for Ns of 1500 or greater }	0.10	0.36
Double volute	0.11	.05	.10
Concentric and modified concentric	.15	.15	.15
Vaned diffuser in volute	.15	.07	.15
Vaned diffuser in concentric	.07	.03	.15

Where:
Ns = specific speed at the best efficiency point.

(2) Effects of thrust reversal where conditions which might cause thrust reversal exist.
(3) Effects of roll, pitch, list and trim of the ship (including gyroscopic effects) shall be considered in the bearing design.
(4) Bearing clearance ratios, L/D ratio and characteristic numbers, lubricant flow, and temperature rise (water-lubricated bearings).
(5) Minimum film thickness required and calculated under maximum radial and axial load conditions over the range of specified pump operating conditions (water-lubricated bearings).

(c) Analysis of driver bearings for close-coupled or rigidly coupled units.
(d) Calculations of the worst case cumulative effect of dimensions and tolerances for parts and locations. These calculations shall demonstrate:

(1) There will no interferences between rotating and stationary parts as-assembled.
(2) There will be no interference between rotating and stationary parts and hence no reduction in life of wearing parts, when these worst case cumulative effects are combined with operational, environmental, and thermal loading.

(e) Casing design (submarine sea water pumps only). In the design of submergence pressure boundaries in sea connected pumps for submarine application, full consideration shall be given to the cyclic nature of the submergence pressure, external nozzle loading, and liquid temperature. The design of the pressure containing parts shall be in accordance with the following basis:

(1) Attached system piping size and material shall be as specified (see 6.2.1).
(2) Pressure and temperature transients as specified (see 6.2.1).

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(3) The average flow velocity in the casing shall be calculated in feet per second at rated flow. This velocity shall be shown for the narrowest and widest sections of the casing.

(f) Calculations demonstrating that the gland leak-off collection and drainage features meet the flow capacity requirement of 3.3.8.3.1.

(g) For surface ship main sea water pumps, casing stress analysis shall be performed and supplied demonstrating compliance with 3.3.3.3.

(h) Calculations demonstrating that assemblies that use springs, such as some mechanical seals and labyrinth seals, shall function as intended given the tolerances on the spring and the tolerances of the parts that restrain the two ends of the spring. As a minimum, a tolerance stackup analysis shall be provided calculating the minimum and maximum valves for working length and the minimum and maximum spring force (see 3.6.1.2.3.3) at those lengths. The basis for believing the assembly will perform properly with those spring forces shall be stated.

(i) Calculations demonstrating that the shaft pressure break device meets the maximum flow requirements of 3.3.8.3 for worst case tolerances. Also, a curve shall be provided, calculated in the same manner, showing how the flow rate increases as the critical clearances increase. This curve shall extend out to a flow rate of 100 gal/rein and shall be for Navy use during shop life.

3.9 Design record. A design record shall be submitted to the design review agency concerned in accordance with the data ordering document included in the contract or order (see 6.2.2). This record shall be completed except as noted in 3.9.2, and forwarded prior to unit delivery or at a time agreed to by the design review agency.

3.9.1 The design record shall contain the following:

(a) The design and calculations which were submitted for manufacturing release, except as modified to resolve comments at time of manufacturing release.

(b) Revised calculations in those areas of design where the original calculations must be modified as a result of an approved design change.

(c) Approved design evaluation test reports, including resolution of comments, the approval action and any special testing which may be performed to verify new design features,

3.9.2 Changes in pump design approved subsequent to issue of the design record when required, or approved deviations in the design evaluation test unit from the design described in the design record, shall be submitted as as addendum to the design record. The addendum shall include applicable revised drawings.

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3.10 Technical repair standard. A technical repair standard (TRS) shall be provided upon initial introduction of a new equipment or introduction of an equipment that has an approved TRS if major or minor configuration changes have been approved that would effect the adequacy of the standard for use in new equipment repairs (see 6.2.2).

3.10.1 TRS technical content shall include sufficient technical details to enable a repair, maintenance, or overhaul activity to restore the equipment's dimensions, clearances, and tolerances such that the equipment is capable of performing its function as originally specified and is capable of being logistically supported by the DOD logistics support system.

3.10.2 The TRS format and content shall conform to DOD-STD-2147 and shall be approved and validated in accordance with this standard.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the contractor is required to maintain, as provided in the inspection clause of the contract or order, shall be in accordance with MIL-I-45208 (see 6.2.1). The completed parts examination (see 4.1.1.1) shall fulfill the completed articles inspection requirements of MIL-I-45208 (see 6.2.1).

4.1.1.1 The inspection system shall include a completed part examination procedure whereby each functionally and dimensionally critical part and assembly, including the final product, is examined on completion in accordance with a completed part examination check list accompanying each part or assembly (see 4.5).

4.1.1.2 A copy of the completed part examination check list for the completed pump, identified by manufacturer's serial number, shall be furnished the contracting activity (see 6.2.2) in duplicate for each pump. On these copies the contractor's quality control manager shall certify that the inspection requirements specified herein have been fulfilled and that the unit is in accordance with the approved drawings and the applicable specifications.

4.1.2 Variations. No variations or deviations from approved design drawings in 3.6.1.2 shall be permitted.

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4.1.3 Mercury prohibition. pumps, drivers, and auxiliary parts furnished by the pump contractor shall be free of mercury contamination and free of functional mercury. During the manufacturing processes, checks, examinations and tests, the product being offered for acceptance shall not come in direct contact with mercury, any of its compounds, nor with any mercury containing device, such as gages and thermometers. The manufacturer shall certify that the product, when shipped, is free from mercury contamination.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed at a laboratory satisfactory to NAVSEA on each class. Either horizontal or vertical units may be submitted. The several first article tests on one pump design may be conducted concurrently, if practicable. The test shall fully establish that the product has the reliability and performance capability specified herein. A first article test report shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2). The following tests shall be performed:

(a) performance evaluation tests specified in 4.6.6 through 4.6.6.3 and the examination specified in 4.5. Performance evaluation tests shall be conducted on one pump of each type, design and size on each contract or order (see 4.6.6).
(b) Design evaluation tests as specified in 4.6.7 through 4.6.7.6. Design evaluation tests shall be conducted on one pump of each type, design and size (see 4.6.7).

4.3.1 First article inspection shall include all sizes of each type and both horizontal and vertical units.

4.3.2 The contractor of a specific class pump shall submit design data and calculations, as specified in 3.8 to the Design Review Activity for review and approval as a prerequisite to the performance of the first article tests specified herein. The design record, as specified in 3.9, shall be submitted upon completion of the first article tests.

4.4 Quality conformance inspection. Each pump (and subcomponents of each pump, if applicable) shall be examined as specified in 4.5 and tested as specified in 4.6.1 through 4.6.5. Quality conformance inspection shall be fully documented by a prescribed test procedure and a written record of the performance and findings of each test (see 6.2.2). A compilation of the documentation of all the quality conformance tests performed on each pump shall be furnished in duplicate (see 6.2.2).

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4.5 Examination. Pumps shall be examined in accordance with a completed part examination check list accompanying each part or assembly. The examination shall be performed by inspection personnel not engaged in the fabrication and assembly of the part. The check list for each part shall include accurate part identification and shall list the specific attributes which determine the part's acceptability. The check list shall be filled in during the time of examination. The check list shall be arranged in tabular form, calling forth separately the following examination assignments, each accompanied by the specific attributes of acceptability which shall be checked or measured, and the findings recorded:

(a) Dimensions. They shall be as indicated on the approved design drawing. Rotating parts shall be examined for balance and maximum permissible run-out as indicated on the drawings.
(b) Materials. Visual examination for compliance with approved design drawings. Any appearance of nonconformance shall be further verified by appropriate tests.
(c) Cleanliness. Examination for cleanliness shall apply equally to parts examination prior to assembly as well as subassemblies and the completed product.
(d) Missing parts. This examination applies primarily to the completed product prior to shipment.
(e) Assembly. During assembly, positioning tolerances and wear ring/impeller running clearances shall be verified.
(f) Alinement. The equipment and its subassemblies shall be examined to insure that the alinement is as specified.
(g) Adjustment. The adjustment of safety, control, and monitoring devices shall be verified and the settings recorded. Settings which should not be tampered with, once adjusted, shall be tagged with an appropriate precautionary warning.
(h) Preservation. The preservation used, the method of application, and recommended procedure for removal of preservative prior to placing the unit in service shall be stated on a supplement sheet and attached to the check list.

4.5.1 A copy of the completed part examination check list for each completed pump, identified by manufacturer's serial number, shall be furnished (see 6.2.2). On these copies, the contractor's quality control manager shall certify that the inspection requirements specified herein have been fulfilled and that the unit is in accordance with the approved drawings and the applicable specifications.

4.6 Tests. The contracting activity and Government representatives shall have the right to examine the facilities at the contractor's plants and at the contractor's subcontractor's plants, and to witness all tests specified herein. Failures, deficiencies, and discrepancies revealed during the performance of the specified tests and the corrective measures taken, shall be recorded and fully documented in the applicable test records and test reports (see 6.2.2). After correction of deficiencies, tests shall be repeated to the extent necessary to ascertain acceptability for the modified pump. Major failures indicative of the design deficiency (as distinguished from shop error or faulty workmanship) shall be reported to the design review agency before a correction is made. Except where specifically required herein to be tested with actual device, tests of turbine driven pumps may be conducted using a substitute driver provided the actual device separately passes the tests required by its specification.

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4.6.1 Hydrostatic pressure test. All pressure boundary parts shall be tested hydrostatically to a pressure one and one-half times the maximum discharge pressure at maximum submergence, but in no case less than 50 lb/in^2. The hydrostatic test pressure shall be maintained for at least 30 minutes or longer as necessary for examination of entire casing.

Acceptance criteria: The pump shall exhibit no leakage through the pressure boundary material or joints. The leaking rate through a mechanical seal shall not exceed 5 drops per minute.

4.6.2 Over-speed test. Each variable speed driver capable of overspeeding shall be over-speed tested in accordance with the specification applicable to the driver.

4.6.3 Mechanical soundness and capacity test. This test shall be conducted, recorded, and reported in accordance with the centrifugal pump Rating Standard and Test Code of the Hydraulic Institute, to the extent that these standards are applicable and are not in conflict with the requirements specified herein. The test report for each pump shall include the following as a minimum: (a) identification of the major pump components (pump, gear assembly, driver) by manufacturer's drawing number and serial number that were tested; (b) identification of the diameter of the impeller(s) tested in the pump; (c) a dimensioned sketch of the test loop showing location of the pump, location of all instrumentation, distance (vertical and along the pipe axis) from the suction and discharge gage taps to the pump suction and discharge flanges, vertical distance from the gage(s) to the elevation datum to which they are calibrated, azimuthal location of the gage taps on the pipe circumference, and location and orientation of any elbows in the pump suction piping; (d) the test loop water temperature during the test; (e) a list of the test instruments including date of last calibration, advertised accuracy, size (e.g., 0.25 lb/in^2) of the smallest graduation on the readout scale, range of the readout scale (e.g., 0-100 lb/in^2), and unit (e.g., lb/in^2) of measurement including the water temperature the gages are calibrated for if a gage is calibrated in feet of water rather than in lb/in^2; (f) the data sheets of all recorded data, with the unit of measurement identified for all data; (g) a sample calculation of each type of calculation converting the raw data to specified conditions and showing the conversion in sufficient detail to permit an independent reviewer to verify the calculations, including all temperature and density corrections; (h) a COPY of the specific table of water properties used in the calculations and a reference to the source of that table; and (i) a plot of the measured head-capacity curve, corrected to the specified operating conditions following the method described in the sample calculations. The test shall be performed as follows:

(a) Operate the pump and its driver, if motor driven, continuously at the maximum rated speed and capacity, with the pumped fluid at ambient temperature until bearing temperatures (not including water-lubricated bearings) stabilize. Stabilization is defined as three consecutively recorded readings taken over intervals of at least 15 minutes that fall within 3°F band when adjusted for ambient. The three consecutive readings shall not be constantly rising. The pump operation shall be monitored for proper functioning of safety devices, bearing lubrication, and for smooth running.

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Acceptance criteria: Unit operation shall be without heating and shall be free of abnormal vibrations and noises. Oil temperature rise in force-feed lubricated bearings shall not exceed 50°F with inlet cooling water to the oil cooler at 85°F. Controls and attached instruments shall function as specified and are in calibration. There shall be no abnormal leakage of water or oil.

(b) Operate the pump at the maximum rated speed with the pumped fluid at maximum normal temperature from recirculation flow as close to free delivery as practicable and with the minimum specified suction pressure prevailing. The unit shall be operated at seven or more test points throughout the full operating range to establish accurately the head-capacity curve at maximum rated speed. The unit shall be operated at each test point until the test values being measured stabilize. In addition to the above testing, determine the required net positive suction head at the maximum rated speed at maximum rated flow rate.

Acceptance criteria: The pump shall deliver the rated capacity and head from the head at the corresponding capacity on the head-capacity curve at maximum rated speed established in the performance test of 4.6.6.1. If more than one performance test has been performed for a given pump design for use in a given ship class, then the head capacity curve established in the first performance test is the one to which the preceding sentence refers. The head-capacity curve at maximum rated speed shall satisfy the specified requirements. The total head at all capacities from O to 120 percent of rated capacity on the curve shall not deviate by more than plus or minus 2.5 percent of rated head under 500 feet. In no case shall the pump deliver less than the rated head. The required net positive suction head shall not exceed the minimum net positive suction head available as specified in 6.2.1. Pumps originally qualified in accordance with earlier revisions of this specification shall continue to meet the head repeatability percentage requirements of the applicable earlier revision, not the above plus or minus 2.5 percent requirement in order to maintain consistency with the original design of the shipboard piping system.

(c) Operate the electric motor-driven unit for a minimum of 1 minute in reverse rotation at maximum rated speed.

Acceptance criteria: The unit shall not be damaged by the reverse rotation test. Conformance to hydraulic and noise requirements shall be demonstrated subsequent to the reverse rotation test.

4.6.4 Noise tests. Airborne and structureborne noise tests when specified (see 6.2.1) shall be conducted and reported in accordance with MIL-STD-740. Noise test details, instrumentation, and testing techniques identified in MIL-STD-740 shall be submitted to the contracting activity prior to testing for approval. Noise tests shall be performed with the driver furnished with pump, and tests shall be conducted on all units. Airborne noise tests shall be conducted on the lead unit only on each contract or order.

Acceptance criteria: The unit shall meet the noise level limits specified in 6.2.1.

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4.6.5 Non-destructive inspection and ultrasonic inspection. All non-destructive inspection, all welding, and all inspection of welds shall comply with MIL-STD-278 and MIL-STD-271.

4.6.6 Performance evaluation tests. One pump of each type, design, and size on each contract or order shall successfully undergo the performance evaluation tests to establish the complete pump performance map and to ascertain compliance with the specified performance requirements. Motor driven units shall be tested with the motor to be furnished with the pump. A substitute driver may be used for turbine driven pumps. When pumps of identical type, design, and size are being produced under two or more contracts within a 2-year period, only one series of performance evaluation tests shall be required for that production 'run. The following tests shall comprise the performance evaluation tests:

(a) Performance test (see 4.6.6.1).
(b) Mechanical shaft seal and pump alinement procedure test (see 4.6.6.2).

4.6.6.1 Performance test. The performance tests shall be conducted, recorded, and reported in accordance with the requirements of the test specified in 4.6.3, except that, in addition, a full performance map shall be established. The proposed test procedure for this test shall be submitted to the contracting activity for approval approximately at the time of drawing submittal. Test data shall be converted to specified operating conditions for plotting of all performance curves. The performance curves shall be determined at maximum and minimum operating speed for variable speed pumps and at each operating speed for multi-speed pumps. A full net positive suction head (NPSH) curve is required on lead production unit and on subsequent units the NPSH shall be determined at design rated capacity only. The following curves shall be established:

(a) Capacity versus total head.
(b) Capacity versus pump efficiency.
(c) Capacity versus brake horsepower.
(d) Capacity versus net positive suction head required.

Acceptance criteria: The acceptance criteria specified in 4.6.3(a) shall be met. The performance map shall exhibit the specified pump performance characteristics. The net positive suction head required shall not exceed the minimum suction head available specified in 6.2.1. Controls and safety devices shall function reliably as intended throughout the full operating ranges of capacity and speed.

4.6.6.2 Mechanical seal and alinement test. Mechanical seal and alinement test shall be as follows:

(a) The mechanical seal shall be installed and alined precisely in accordance with the procedures stated in the proposed pump technical manual. Packing shall not be installed during this test. The amount of force required to axially position the mechanical seal onto the impeller hub, the shaft sleeve or whatever it rests on in the assembled condition shall be measured and recorded. If the mechanical seal includes an adhesive or setting compound on the elastomer that seals between the mechanical seal and the impeller hub or shaft

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sleeve, and if there is a maximum allowed time between initial wetting of that compound and final positioning of the mechanical seal in the pump, then do not perform that final positioning until the maximum allowed time has clasped.

(b) After installation, the mechanical seal shall be run-in and the pump operated for at least 4 hours at maximum rated speed with the pumped fluid at maximum operating temperature and with pump suction pressure as high as feasible without requiring pump test loop modifications. The seal leak rate shall then be measured and recorded. Also, the dynamic balance shall be measured and recorded using a vibration indicator of the neon tube or similar type capable of measuring vibrations of 0.001 inch or less. During subsequent pump disassembly, the final axial position of the mechanical seal shall be measured and recorded if the pump design permits.

Acceptance criteria: Mechanical seal leakage shall not exceed 5 drops per minute. The pump and drive shall exhibit no abnormal noises or roughness of operation. Vibration shall not exceed the limits specified in MIL-STD-167-1, type II, measured on the bearing caps. The mechanical seal shall have been positioned at the axial position recommended by the mechanical seal vendor to generate the proper amount of mechanical seal spring force. The mechanical seal vendor shall provide in writing his recommendation regarding the axial position in his concurrence that the mechanical seal alinement test results demonstrated proper positioning of the seal. If it was not practicable to measure the axial position of the mechanical seal, then the seal vendor shall concur that the recorded force used to position the seal during assembly was acceptable and shall state the technical basis for considering that force acceptable. A copy of these mechanical seal vendor documents shall be provided as part of the test report.

4.6.6.3 Packing and alinement test. The mechanical seal shall not be installed during this test. The pump shall be packed and alined precisely in accordance with the procedures stated in the proposed technical manual. The packing shall then be run in the pump and operated for at least 4 hours at a maximum rated speed with the pumped fluid at maximum operating temperature and with pump suction pressure as high as feasible without requiring pump test loop modifications. The time of day of each packing leakage measurement and of each packing adjustment shall be recorded. The dynamic balance shall be checked with a vibration indicator. After the tests, the packing and shaft sleeve shall be inspected and the results recorded. If the rated maximum operating temperature exceeds 200°F and the normal operating temperature is less than or equal to 200°F, the packing and alinement test shall be repeated satisfactorily with the pumped fluid at normal operating temperature.

Acceptance criteria: The stabilized packing leak rate, defined as that measured at least 30 minutes after the last packing adjustment, shall be within the limit recommended by the pump vendor. Vibration shall not exceed the limits specified in MIL-STD-167-1, type II, measured on the bearing caps. There shall be no smoking or scorching of packing and no scoring (vice polishing) of the shaft sleeve.

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4.6.6.3.1 Test reports for the performance evaluation tests shall be compiled into a single document. The document shall identify the pump by manufacturer's serial numbers of pump and driver, the contract or order number under which the tests were performed, the ship or class of ships for which the pump is intended if known, and the letter and date of test procedure approval. The test compilation document shall be distributed as specified (see 6.2.2).

4.6.7 Design evaluation tests. One pump of each type, design, and size complete with the driver and all appurtenances and controls shall successfully undergo the design evaluation tests specified herein. Design changes which in the opinion of the design review agency may detrimentally affect the reliability of a previously tested and accepted pump design shall be cause to require new design evaluation tests in part or in full. It shall be at the discretion of the design evaluation tests when an accumulation of several design changes, each of itself apparently not significant to pump reliability, collectively considered render suspect the validity of any one or all or the previously performed design evaluation tests. An endurance test need not be performed provided that the vendor can demonstrate to the satisfaction of the design review agency concerned that the pump to be offered has been proven in previous Navy shipboard service under conditions equally as strenuous as those specified in the ordering data. The proposed design evaluation test procedures shall be submitted for approval to the contracting activity prior to performing the tests. The design evaluation tests shall consist of the following:

(a) Endurance test (see 4.6.7.1).
(b) Shock test (see 4.6.7.2).
(c) Inclined operation test (see 4.6.7.3).
(d) Vibration tests (if specified (see 6.2.1)) (see 4.6.7.4).
(e) Experimental stress analysis (see 4.6.7.5).

4.6.7.1 Endurance test. The pump shall be operated for a period of not less than 500 hours of actual running time with a minimum of 100 starts to ascertain reliability of performance and operation.

4.6.7.1.1 Prior to commencement of the endurance test and immediately after completion of the 500-hour operating run, the pump shall be disassembled to the extent necessary and the critical dimensions and running clearance of parts subject to wear, erosion and derangement, shall be measured, calculated and recorded. Components such as pump impellers and casings subject to erosion, corrosion, cavitation, and wear, the effects of which are not subject to routine measurement, shall be listed in the inspection record and after completion of the test the condition of each component determined and recorded.

4.6.7.1.2 During the initial and final hours of the endurance test run, noise and performance tests in accordance with 4.6.4 and 4.6.6.1 shall be performed to determine the changes in the pump's performance characteristics and noise signature. Vibration measurements shall be taken at the bearings caps or housings of the pump and driver at equal speeds during the initial and final capacity test to determine the changes in mechanical operation.

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4.6.7.1.3 The endurance test shall not be continuous but shall be interrupted by at least three rest periods of a minimum of 8 hours each. A minimum of 100 starts at full line voltage shall be performed during the course of the test. The remainder of the endurance tests shall be run at maximum rated speed and within plus 20°F, minus O°F of maximum specified liquid temperature. The pump shall be operated at one-third, two-thirds, and rated capacity in approximately equal time intervals. Operations at rated capacity shall be at minimum specified net positive suction head available, or maximum specified auction lift or vacuum, as applicable.

4.6.7.1.4 The pump shall be monitored during the endurance test to accurately record the conditions of operation, the capacity delivered, the total head developed, the speed at which operated, and the general performance observed. Data shall be collected and the pump inspected at least twice per day of operation. For each periodic inspection, in addition to all measured data, the record shall indicate the following:

(a) Seal leakage rate and the amount of gland leakage collected and drained by the collection area.
(b) The conditions of the bearings (by audible noise; by feel; and by bearing temperature via probe if the design includes provisions for a probe, otherwise via a surface pryometer on a normally exposed surface; no disassembly required).
(c) The airborne noise level (normal-abnormal).
(d) The vibration level (normal-abnormal).
(e) The smoothness of operation (normal-abnormal).
(f) All other abnormal findings.
(g) All adjustments made.
(h) Changes made in the conditions or method of operation.

4.6.7.1.5 Endurance test acceptance criteria. Endurance test acceptance criteria shall be as follows:

(a) Head-capacity curve at maximum rated speed after 500 hours of pump operation shall conform to the specification requirements and shall show no abnormal deviations from the curve before the 500-hour test.
(b) Unit performance and operation after 500 hours of operation shall be unchanged and normal and meet all specification requirements.
(c) Unit operation at the end of the endurance test shall be smooth and shall exhibit noise and vibration levels that are normal and in conformance with the specification.
(d) Lubrication shall have remained satisfactory throughout the test period. Bearing temperatures shall have remained normal and shall be consistent with their respective bearing clearances and oil and grease limitations.
(e) Leakage rate from a mechanical shaft seal shall not exceed five drops per minute.
(f) Running clearances shall be normal.

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(g) Components subject to attack from corrosion, erosion, cavitation, etc. shall be in a condition commensurate with 500 hours of service.

(h) Wear rates for wearing parts, critical for proper operation, shall show a rate of wear for the test period that shall be consistent with the specified design life requirements.

(i) No failures shall have occurred throughout the 500-hour test.

4.6.7.1.6 Post endurance test procedures. The unit subjected to the 500-hour endurance test shall be restored to the as-new condition by replacement of all parts worn beyond the as-new design tolerances. The restored unit shall successfully pass the tests specified in 4.6.1, 4.6.3(a), and 4.6.4, if applicable. The quality conformance test documentation shall indicate that the unit was subjected to the endurance test and subsequently restored, tested, and that it shall be certified as fully conforming to the specification for unrestricted service.

4.6.7.2 Shock test. The pump shall undergo a shock test to ascertain that the pump has the necessary shock resistance. The shock test shall be performed in accordance with MIL-S-901, grade A, and the specific shock test requirements (see 6.2.1).

4.6.7.2.1 Before and after shock test, the pump and driver and other components susceptible to internal derangement shall be disassembled to the extent necessary and the critical dimensions and running clearances measured, calculated, and recorded. During this disassembly, the critical components and assemblies subject to shock damage and derangement shall be identified and listed in the Inspection record and after completion of the test, the condition of each component and assembly shall be determined and recorded. Shafts, impellers, and turbine rotors, motor rotors, and reduction gears shall be inspected by one of the applicable non-destructive test procedures, other than radiography, specified in MIL-STD-271.

4.6.7.2.2 Before and after the shock test, tests in accordance with 4.6.3 shall be performed to determine the changes in performance characteristics of the pump. Vibration measurements shall be taken at the bearing caps or housings of the pump and driver at equal speeds during the initial and final capacity test to determine the changes in mechanical operation.

4.6.7.2.3 The unit shall be mounted on the shock machine or barge essentially identical to the actual shipboard installation. The contracting activity will furnish the contractor a drawing of the shipboard mounting arrangement and foundation's stiffness. Horizontal pumps, when tested in the inclined position on the medium weight shock machine, shall be oriented so that the direction of shock is perpendicular to the axis of the pump rotation. The pump shall be in operation during the first, third, and fifth blows of the shock test on the floating barge. Variable speed pumps shall be operated at the minimum speed and pressure required to insure lubrication of bearings and wearing parts. Other pumps shall be operated at the lowest rated speed.

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4.6.7.2.4 The pump shall be carefully observed during each shock blow and thoroughly examined visually after each blow. After each blow, the unit shall be operated at as close to maximum rated speed as possible and checked for abnormal noises and vibrations and proper functioning of controls. Turbine driven pumps may be air driven. Tightening of bolts during shock tests will not be permitted. If any bolt loosens during the test, the equipment manufacturer shall provide a corrective procedure which must be approved by the design review agency.

4.6.7.2.5 Shock test acceptance criteria is as follows:

(a) There shall be no breakage of parts, including mounting bolts.
(b) There shall be no distortion or derangement of any part which would render the unit incapable of performing as specified.
(c) The amplitude of vibration after test at maximum rated speed shall be less than twice the amplitude measured at the same speed before the test.
(d) Adequate lubrication to all bearings shall be maintained.
(e) Critical dimensions and running clearances have been maintained.
(f) There shall be no significant change in the head-capacity curve.

4.6.7 .2.6 Post shock test procedure. The shock tested unit, if it is to be supplied under a contract or order, shall be restored to the as-new condition by replacement of all parts damaged or distorted beyond the as-new design tolerances. Rolling contact bearings shall be replaced regardless of condition. The shock tested rolling contact bearings shall be destroyed. The restored unit shall successfully pass the tests specified in 4.6.1, 4.6.3(a), and 4.6.4, if applicable. Quality conformance test documentation shall certify that the unit was subjected to the shock test and subsequently restored, tested, and inspected in accordance with this specification. A completed parts examination check list shall be supplied (see 6.2.2) and shall identify the parts which were replaced (such as the bearings) and shall certify that the unit fully conforms to the specifications for unrestricted service.

4.6.7.2.7 Unless otherwise specified (see 6.2.1), pump units shall be shock tested with drivers. Flexible coupled pumps shock tested with one driver will not be required to be shock tested again when supplied with a different driver of equal or less weight. Prime movers are subject to shock tests in accordance with applicable equipment specifications.

4.6.7.3 Inclined operation test. Vertical pumps shall be operated for not less than 30 minutes inclined at an angle from the normal equal to the combination of the maximum permanent list and trim as specified (see 6.2.1). Horizontal pumps shall be tested in accordance with the inclined operation test specified for the driving unit. Operation in the inclined position shall be as close to rated speed and capacity as practicable. The performance point shall be indicated in the test report. The pump shall meet specified performance requirements (see 3.2) without damage while operating in the inclined position.

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4.6.7.4 Vibration test. When specified (see 6.2.1), one pump of a quantity of identical pumps being supplied shall successfully undergo a vibration test in accordance with the requirements of MIL-STD-167-1, type I, and as supplemented in the contract or order. The vibration test need not be repeated on subsequent contracts or orders for pumps of identical design to those previously tested, provided the previous tests included the frequencies specified (see 6.2.1).

4.6.7..4.1 The unit shall be mounted on typical shipboard foundations during the vibration test or the shipboard mounting arrangement shall be simulated in spring mass characteristics except where this mounting arrangement causes the largest test table capacity to be exceeded. Inability to vibration test the unit because of excessive weight or size shall not release the contractor from furnishing equipment which can withstand the specified vibration inputs. Vibration test acceptance criteria shall be in accordance with MIL-STD-167-1, type I.

4.6.7.5 Experimental stress analysis. For submarine sea-connected pumps subject to full submergence pressure, the structural adequacy of one pump casing of each design shall be verified by an experimental stress analysis.

4.6.7.6 Design evaluation test documentation. The test reports for the design evaluation tests shall be prepared in accordance with accepted engineering practice. The test arrangement and procedure, the test events, and test instruments used, the measured data, and the effects, results, and observations shall be accurate and complete and shall be presented in a professional manner and in a usable durable form to the design review agency for approval. The report shall include a definite statement regarding conformance to the acceptance criteria specified herein and suitability of the unit for its intended application. After approval, test reports shall be distributed as specified in 6.2.2.

4.7 Packaging inspection. Sample packages and packs and the inspection of the packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The preparation of delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Technical data. Unless otherwise specified (see 6.2.1), technical data depicting the preservation, packaging, and transportation of support data shall be prepared by the contractor and submitted to the contracting activity (see 6.2.2).

5.2 Preservation-packaging, packing and marking. Pump units and accessories shall be preserved-packaged level A or C, packed level A, B, or C as specified, (see 6.2.1) and marked in accordance with MIL-P-16789.

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6. NOTES

6.1 Intended use. Pumps covered by this specification are intended for pumping sea water through the condenser for condensing steam.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) General design required (see 3.2).
    (1) Space and weight requirements.
    (2) List, pitch, roll, trim, and other ship attitude conditions (see 3.2, 3.2.7, and 4.6.7.3).
    (3) Environmental vibration frequencies (see 4.6.7.4).
    (4) Airborne, structureborne, and fluidborne noise requirements (see 3.2.5 and 4.6.4).
    (S) Ambient temperature, normal, maximum, and minimum.
(d) Performance characteristics (see 3.2, 3.3.2.1, and 3.8.4).
    (1) Rated capacity and range of capacity.
    (2) Rated total head at rated capacity.
    (3) Fluid to be pumped.
    (4) Specific gravity of fluid.
    (5) Temperature of fluid, normal, maximum, and minimum.
    (6) Suction head available, normal, maximum, and minimum.
    (7) Casing design and test pressures.
    (8) Special performance requirements, if any (e.g., maximum shutoff head).
(e) Type of driver.
(f) Ordering data for driver (see 6.1 of applicable specification; MIL-M-17413 or MIL-M-17060 for motors; MIL-T-17523 for turbines; for submarine motors list complete motor specifications) (see 3.2.14).
(g) Ordering data for electrical controllers, if required (see 6.1 of MIL-C-2212) including variable speed controllers.
(h) Whether shock mounts are to be used (see 3.2.3.2).
(i) Horizontal or vertical mounting (see 3.2.6.1).
(j) If threaded fastener lengths are to be other than specified (see 3.2.10.2).
(k) Additional requirements for valves, flanges, fittings, and bolting for pipe connections (see 3.2.18.1).
(l) If coupling is other than specified (see 3.3.1).
(m) If bilge connection requirements are other than specified (see 3.3.2.3, 3.3.2.4, and 3.3.3.2).
(n) Casing nozzle loadings (see 3.3.3.3).
(o) If suction and discharge connection requirements are other than specified (see 3.3.4.1).
(p) Method for sealing shaft gland (see 3.3.8.1).
(q) Material for labryinth seal floating ring (see table X).
(r) Material for labryinth seal stationary ring (see table I).
(s) Number of sets of special tools (see 3.4).

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(t) Drawings required (see 3.6.1.2.6).
(u) Manuals required and whether complete information for electrical equipment is required (see 3.7.3 and 3.7.6).
(v) Information required for stress analysis of submarine seawater pumps (see 3.8.4(e)).
(1) Pressure and temperature transients.
(2) System piping size and material if rigidly attached.
(w) Inspection system (see 4.1.1).
(x) Mercury prohibition (see 4.1.3).
(Y) Shock test requirements (see 4.6.7.2).
(z) If pump is to be shock tested without driver (see 4.6.7.2.7).
(aa) Vibration test requirements (see 4.6.7.4).
(bb) Preservation and levels of packaging and packing (see 5.2).
(cc) When packaging technical data is not required (see 5.1).
(old) Identification of contracting activity and design approval agency (see 6.4).
(ee) Ship type in which units are to be installed.
(ff) Designation of manufacturer (pump or driver) responsible to the purchaser for compatibility of pump and driver and operation of the assembled units.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option	Deliverable to
3.2.2.3, 3.2.7, 3.8, 3.9, and 4.6.7.5	Diagrams, calculations and stress	UDI-E-23253	---	1,2,3,4,5,6,7
Table I, footnote 6, 4.1.1.2, 4.1.3, 4.5.1, and 4.6.7.2.6	Certificate of compliance	DI-E-2121	----	1,2,3,4,5,6,7
3.6	Drawings, engineering, and associated lists	DI-E-7031	Level 2 or 3 Design activity designation - Contractor Drawing No. - Contractor Delivery of hard copy - Contracting Activity	1,2,3,4,5,6,7

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Paragraph no.	Data requirement title	Applicable DID no.	Option	Deliverable to
3.7.1	Manual, technical, preliminary	DI-M-2043	Type I of MIL-M-15071	1,2,3,4,5,6,7
3.7.1	Manual, technical standard	DI-M-2044	Type I of MIL-M-15071	1,2,3,4,5,6,7,8
3.10	Technical/maintenance overhaul and repair standards	UDI-L-26481	---	1,2,3,4,5,6,7,8
4.3	First article inspection report	DI-E-4902	----	1,2,3,4,5,6,7
4.4 and 4.6	Procedures, test	UDI-T-23732	----	1,2,3,4,5,6,7
4.4, 4.6.6.3, and 4.6.7.6	Reports, test	DI-T-2072	10.1.b	1,2,3,4,5,6,7
5.1	Preservation, packaging, and transportation support data	UDI-P-23508	----	1,2,3,4,5,6,7
6.5	Supplementary provisioning technical documentation (SPTD)	DI-V-7000	----	2,8
	Manufacturer or commercial manual (provisioning)	DI-V-7001	----	2,8
	Provisioning parts list (PPL)	DI-V-7002	----	2,8
	Short form provisioning parts list (SFPPL)	DI-V-7003	— --	2,8
	Interim support items list (ISIL)	DI-V-7006	-—-	1,2,3,5,6,7,8

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CODE No.	ACTIVITY
	Naval Sea Systems Command Washington, DC 20362
1	56Y21 (2 Copies)
2	08L (Propulsion plant pumps in submarines and nuclear surface ships only)
3	Contracting Activity
4	Design Review Agency
5	Cognizant Supervisor of Shipbuilding
6	Shipbuilder
7	Defense Contracting Administration Service Office
8	Ships Parts Control Center Mechanicsburg, Pennsylvania

NOTE: Numbers 3 and 6 may be the same activity and only one copy thus may be required.

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. As used herein, “contracting activity” refers to the activity purchasing the equipment covered by this specification. The contracting activity may be the Government or a shipbuilder. The “design review agency” refers to the activity responsible for controlling the technical requirements for design and testing of the equipment. The design review agency is generally a Government command or agency such as NAVSEA or an authorized representative, Communication with the design review agency should be handled through the contracting activity.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

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6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5.2 On board. The repair parts listed are shown for guidance only in preparing provisioning technical documentation.

TABLE II. On board repair parts.

Name of part	Quantity per set
Ball bearings Roller bearings Internal water lubricated bearings Coupling bushings, pins, springs, and other wearing parts Felt wiper rings or oil seals Packing, if special Mechanical seal Shaft sleeves	100 percent complete replacement for all installed units per ship.
Sleeve bearings Undersize casing wearing rings Casing liners Couplings, gear type Stuffing box throat bushings Impeller nuts	100 percent replacement for one pump per ship.

6.5.3 Each box containing on board repair parts should contain a list entitled “List of on board repair parts”. The list should be in a format for use on outline drawings and in the technical manuals. It should be not less than nominal 8-1/2 by 11 inches in size.

6.5.3.1 The format of the list should include a heading and columns of data for the items listed. The heading should include titles and applicable entries as follows:

Reproduced from drawing number _____
Number of ships _____
Application _____
Contract number _____
Manufacturer _____
Quantities are for _____ units per ship
The columns shall include:
Piece number _____
Name of part _____
Quantity _____
Drawing number _____
National stock number _____
Additional columns may be used as applicable.

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6.5.3.2 The list should be so treated as to be resistant to oil, water, and fading.

6.5.4 The repair parts requirements for the drivers and accessories should be in accordance with the related equipment specification.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 does not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(project 4320-N176)

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